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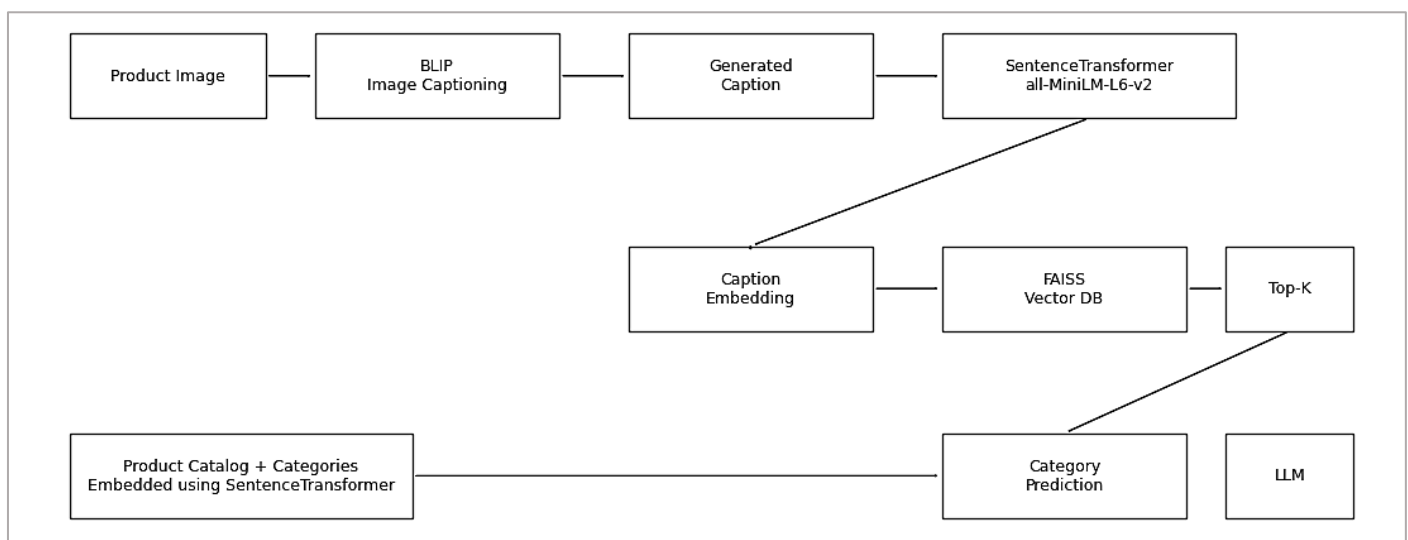
Problem Statement: Imagine you are building an AI system for an e-commerce platform like Amazon or Flipkart. The system should be able to understand a product using image.

Task:

1. **Inputs:** Product Image
2. **Outputs:** Product category, Product description
3. **Model Selection:**
 - a. BLIP (Image Captioning)
 - b. SentenceTransformer(all-MiniLM-L6-v2) – [Mini Language model lightweight version of BERT]
 - c. FAISS Vector DB
 - d. LLM for final description generate.

4. **System Desing (High-Level):**

➤ **Architectural Diagram:**



➤ **Workflow:**

- a) **Product Image:** The input image Uploaded by the customer.
- b) **BLIP Image Captioning Model:** Analyzes the image and generates a textual description of the product.
- c) **Generated Caption:** The text representation of the product generated by BLIP, like "A pair of blue running shoes."
- d) **SentenceTransformer:** Converts the caption into a numerical embedding that captures semantic meaning.
- e) **Caption Embedding:** The vector representation of the product caption used for similarity search.

- f) **Product catalog + Categories:** A database containing product information and category labels. Each product description is also converted into embeddings.
- g) **FAISS VDB:** Stores product embeddings and performs fast similarity search to find the most relevant products or categories.
- h) **Top-K similarity:** Retrieves the k=3 or k = 5 similar products from the catalog based on embedding similarity.
- i) **Category Prediction:** Determines the most appropriate category using the retrieved similar products.
- j) **LLM:** Uses the retrieved information and image caption to generate a detailed product description.
- k) **Final Output:** Returns the predicted product with category and description.

Note:

i. **Why Choose BLIP over CLIP?**

This model is the simplest and most logical choice because the final output requires human-readable description, not just classification. It well in generates text captioning, good for description generation, and also it trained on approx 129M image-text dataset. While CLIP good for generate image embeddings and image classifications.

ii. **Uses FAISS over ChromaDB?**

For Faster, Open-Source, lower memory required and no need to store the metadata.

5. Evaluation: Accuracy matrix, Precision, Recall, F1-Score, BLEU

6. Business Value:

For Sale Person's: Automatic product listing, less effort require.

For Customer's: Better search, Improve satisfaction.

For Stakeholder's: Easy to track growth, easy to manage.

7. Company already used:

AMAZON-Nova: Works for Smart AI agent like navigate web-pages, check inventory, Virtual Try & Visuals, Making interactive shopping experiences. [[reference](#)]

Before Amazon-Nova amazon uses "**Rufus**": A GenAI Shopping Assistant, It trained on Amazon's huge product catalog, customer reviews, web data, and Q&A to answer customer's questions.